METROPOLITAN LIVER & METABOLIC HEALTH CENTER

Department of Clinical Pathology & Computational Hepatology

ACCREDITATIONS: CAP #8492019 • CLIA #36D2081942
ISO 15189 Certified Diagnostic Protocol
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OFFICIAL CLINICAL TEST REPORT — NON-ALCOHOLIC FATTY LIVER DISEASE (NAFLD / MASLD)

Patient Assessment ID: NAFLD-TEST-COMPLETE-001	**Specimen Type:** Peripheral Venous Blood & Clinical Panel
Demographics: 52 Years • Male	**Collection Date:** 2026-10-23 09:55
BMI & Anthropometrics: 31.5 kg/m² (Obese Class I)	**Reporting Status:** FINAL / CLINICALLY VERIFIED
Referring Physician: Dr. Alex Reed, MD (Consultant Hepatologist)	**Diagnostic Assay:** AI Multi-Biomarker Risk Stratifier v2.0

PRIMARY SCREENING STATUS	**POSITIVE (NAFLD INDICATED)**	**SCREENING CONFIDENCE**	**99.2% (92.5% Specificity Benchmark)**
HISTOLOGICAL SEVERITY GRADE	**MODERATE RISK (GRADE 2)**	**STAGE CONFIDENCE**	**62.4% (Multi-Class Staging)**
ACTIVE AI ARCHITECTURE	**Improved Random Forest Pipeline (250 Trees, Notebook Model)**	**CLINICAL ACTION PATHWAY**	**Active steatohepatitis. Schedule abdominal ultrasound and hepatology consultation.**

Grade 0 (None)	Grade 1 (Mild)	Grade 2 (Moderate)	Grade 3 (Severe)
3.6%	10.8%	**62.4%** • Pred	23.2%

COMPLETE LABORATORY TEST PANELS & PATHOLOGICAL FLAGS

PANEL 1: HEPATIC TRANSAMINASES & FUNCTIONAL LIVER MARKERS	Result	Unit	Reference Interval	Clinical Flag
Alanine Aminotransferase (ALT)	**68**	U/L	7 - 56	**Elevated (H)**
Aspartate Aminotransferase (AST)	**54**	U/L	10 - 40	**Elevated (H)**
AST / ALT De Ritis Ratio (Derived)	**0.79**	Ratio	0.8 - 1.2 (<1.0 typical for NAFLD)	**Normal**
Gamma-Glutamyl Transferase (GGT)	**88**	U/L	9 - 48	**Elevated (H)**
Total Bilirubin	**1.2**	mg/dL	0.1 - 1.2	**Normal**
Platelet Count	**195**	×10³/µL	150 - 450	**Normal**

PANEL 2: LIPID METABOLISM & ATHEROGENIC BIOMARKERS	Result	Unit	Reference Interval	Clinical Flag
Serum Triglycerides	**240**	mg/dL	< 150	**High (H)**
HDL Cholesterol (Cardioprotective)	**38**	mg/dL	> 40 (M) / > 50 (F)	**Low Risk Flag (L)**
LDL Cholesterol (Calculated)	**145**	mg/dL	< 100	**Elevated (H)**
Triglyceride-to-HDL Ratio (Derived)	**6.32**	Ratio	< 3.0 (<3.5 Non-Insulin Resistant)	**High Atherogenic Index**

PANEL 3: GLYCEMIC CONTROL & ENDOCRINE MARKERS	Result	Unit	Reference Interval	Clinical Flag
Fasting Blood Glucose (FBG)	**128**	mg/dL	70 - 99 (Normal)	**Diabetic Range (H)**
Thyroid Stimulating Hormone (TSH)	**3.1**	µIU/mL	0.4 - 4.5	**Normal**

PANEL 4: ANTHROPOMETRICS, COMORBIDITIES & LIFESTYLE PROFILE	Result	Unit	Reference Interval	Clinical Flag
Body Mass Index (BMI)	**31.5**	kg/m²	18.5 - 24.9 (Normal)	**Obese Class I**
Type 2 Diabetes Mellitus (T2D)	**Yes / Positive (1)**	Clinical Status	0 = Absent, 1 = Confirmed	**Present / Flagged**
Essential Arterial Hypertension	**Yes / Positive (1)**	Clinical Status	0 = Absent, 1 = Confirmed	**Present / Flagged**

Polycystic Ovary Syndrome (PCOS)	**N/A (Male)**	Clinical Status	0 = No, 1 = Yes, 2 = N/A (Male)	**Not Applicable**
Alcohol Consumption (Clinical Intake)	**Yes / Positive (1)**	Clinical Status	0 = No, 1 = Yes (Qualifies for NAFLD)	**Present / Flagged**
Tobacco Smoking History	**Yes / Positive (1)**	Clinical Status	0 = Non-Smoker, 1 = Smoker	**Present / Flagged**

Statistical Biomarker Drivers: BMI (28.0%) • **ALT** (10.0%) • **TRIGLYCERIDES** (10.0%) • **AST** (8.0%) • **GGT** (8.0%)

COMPREHENSIVE CLINICAL INTERPRETATION & MANAGEMENT PLAN

Clinical Interpretation: Significant biochemical transaminitis and hypertriglyceridemia consistent with active Non-Alcoholic Steatohepatitis (NASH/MASH). Risk of ongoing hepatocellular injury and progressive architectural changes. **Actionable Plan:** Order abdominal ultrasonography, schedule specialist gastroenterology/hepatology consultation, and screen for cardiovascular and metabolic comorbidities.

Electronically Verified & Signed By:
Dr. Alex Reed, MD, PhD, FACG
Consultant Hepatologist & Medical Director
License #MED-8849201 • NPI: 1982740192

Diagnostic Verification Hash:
SHA256:NAFLD-TEST-COMPLETE-001-AUTH-20261023
Laboratory Director Approval: VALIDATED

LEGAL & REGULATORY NOTICE: This clinical summary is generated by an artificial intelligence diagnostic screening prototype designed for academic, research, and non-invasive triage purposes. The findings do not constitute an independent definitive diagnosis. Formal diagnostic confirmation requires professional physician evaluation, abdominal imaging (ultrasonography / FibroScan), or histopathological biopsy where clinically indicated.